ENTECAVIR

Empty stomach

Do not stop suddenly

CHEMICAL NAME 2-amino-9-[(1S,3R,4S)-4-hydroxy-3-(hydroxymethyl)-2-methylidenecyclopentyl]-1,9-dihydro-6H-purin-6-one (monohydrate)

CLASS Guanosine nucleoside analogue (HBV polymerase inhibitor)

TRADE NAMES Baraclude · Entecavir (generic)

THERAPEUTIC USES / INDICATIONS

- Chronic hepatitis B (compensated and decompensated liver disease); first-line.
- HBV prophylaxis during chemotherapy/immunosuppression.

ADMINISTRATION

- **PO** on an empty stomach (at least 2 h after and 2 h before a meal).
- Once daily at the same time.

DOSAGES

- **Treatment-naïve:** 0.5 mg once daily.
- **Lamivudine-refractory or decompensated:** 1 mg once daily.
- **Children ≥2 yr (≥10 kg):** weight-based oral solution.
- Reduce dose or extend interval if CrCl <50 mL/min.

PHARMACOKINETICS

- **Peak:** 0.5–1.5 h. Food delays absorption.
- Minimal metabolism; not a CYP substrate.
- Excreted unchanged in urine. **t½:** 128–149 h.

PHARMACODYNAMICS

- Phosphorylated to active triphosphate → inhibits **HBV DNA polymerase** at priming, reverse transcription and DNA synthesis.
- Potent suppression with high barrier to resistance (in treatment-naïve patients).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, fatigue, dizziness.
- Nausea, dyspepsia.
- Insomnia.

ADVERSE EFFECTS (SERIOUS)

- Severe acute exacerbation of hepatitis B on stopping.
- Lactic acidosis and hepatomegaly with steatosis (rare; ↑ in decompensated cirrhosis).
- HIV resistance if used in untreated HIV co-infection.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- **Caution:** untreated HIV co-infection, renal impairment, liver transplant.

KEY INTERACTIONS

- **Nephrotoxic drugs:** may ↑ levels.
- Few significant interactions.

NURSING MANAGEMENT

- Check HIV status before starting.
- Monitor LFTs, HBV DNA every 3–6 months; renal function.
- Reinforce adherence and empty-stomach dosing.
- Watch for signs of lactic acidosis (malaise, tachypnoea, abdominal pain).
- Hepatitis B vaccination for household contacts.

PATIENT EDUCATION

- Take on an empty stomach at the same time daily.
- Do not stop without medical advice.
- Avoid alcohol; avoid sharing razors/toothbrushes.
- Family members should be vaccinated.

KEY POINTS

- Potent HBV polymerase inhibitor; first-line.
- 0.5 mg daily on an empty stomach.
- Hepatitis flare if stopped.
- Check HIV status first.